

QUANTITATIVE DATA ANALYSIS IN HCI

Dr. Tom Biselli · Department of Computer Science · Aalborg University

23.04.2026

Please open the R script
before the lecture starts
<https://bit.ly/cookie-stats>

What You Will Be Able to Do After Today

OVERVIEW

01

Collect & structure

Explain how quantitative data is collected and structured in HCI studies

02

Hypothesise

Translate a research question into testable H_0 / H_1 hypotheses

03

Describe

Compute and interpret descriptive statistics - and know their limits

04

Choose & test

Select the right statistical test and run it in R

05

Interpret

Read p-values, effect sizes, and post-hoc tests correctly

06

Report

Write up results in APA style from raw R output

Today's Journey

OVERVIEW



Hands-On R code through sections 4 - 7

1

SECTION 1

MOTIVATION

Why quantitative methods in HCI?

1

Motivation

2

Paper Pitch

3

Hypotheses

4

Study Design

5

Descriptive Stats

6

Inference Stats

7

Reporting

8

Exercise

You Already Know Both Sides - Let's Connect Them

MOTIVATION

Qualitative Research

Interviews, think-aloud, ethnography

Rich in context and meaning

Uncovers the 'why' and 'how'

Small N, deep understanding

Not generalisable to populations

Quantitative Research

Surveys, experiments, logs

Measurable, comparable, replicable

Answers 'how much' and 'does it differ'

Larger N, goal: generalisable

Requires careful experimental design

Today's paper uses both: qualitative interviews → design → quantitative evaluation

SECTION 2

PAPER PITCH

2

Tom Biselli, Laura Utz, Christian Reuter (2024)

Supporting Informed Choices about Browser Cookies: The Impact of Personalised Cookie Banners

Proceedings on Privacy Enhancing Technologies (PoPETs)

1

Motivation

2

Paper Pitch

3

Hypotheses

4

Study Design

5

Descriptive Stats

6

Inference Stats

7

Reporting

8

Exercise

The Paper: Supporting Informed Cookie Choices

01 Problem

Cookie banners are often difficult to understand and use dark patterns - they nudge users to accept all cookies and undermine informed consent

02 Approach

User-centred design: 19 semi-structured interviews → requirements → alternative cookie banner → controlled experiment

03 Personalisation

Cookie banner explanation text tailored to users' privacy knowledge level (low / medium / high) via OPLIS questionnaire

04 Evaluation

N = 157 via Prolific, 3 groups, 7 simulated websites, online experiment

Three Experimental Groups - Who Saw What?

CG

Control Group

Standard cookie banner of each website

The banner you normally see - including challenging texts and dark patterns

N = 53

PA

Privacy Assistant

Novel redesigned banner

Neutral, no dark nudges, 3 equal options

Same text for everyone

N = 49

PPA

Personalised Privacy Assistant

Same novel banner as PA + explanation text tailored to user's privacy knowledge (measured via OPLIS score and grouped into low - med - high)

N = 55

Novel Redesigned Cookie Banner

Cookie Assistant

This website wants to use cookies. These can have different functions and may include third-party cookies. You can now decide which cookies the website may use.

▼ What is a cookie?

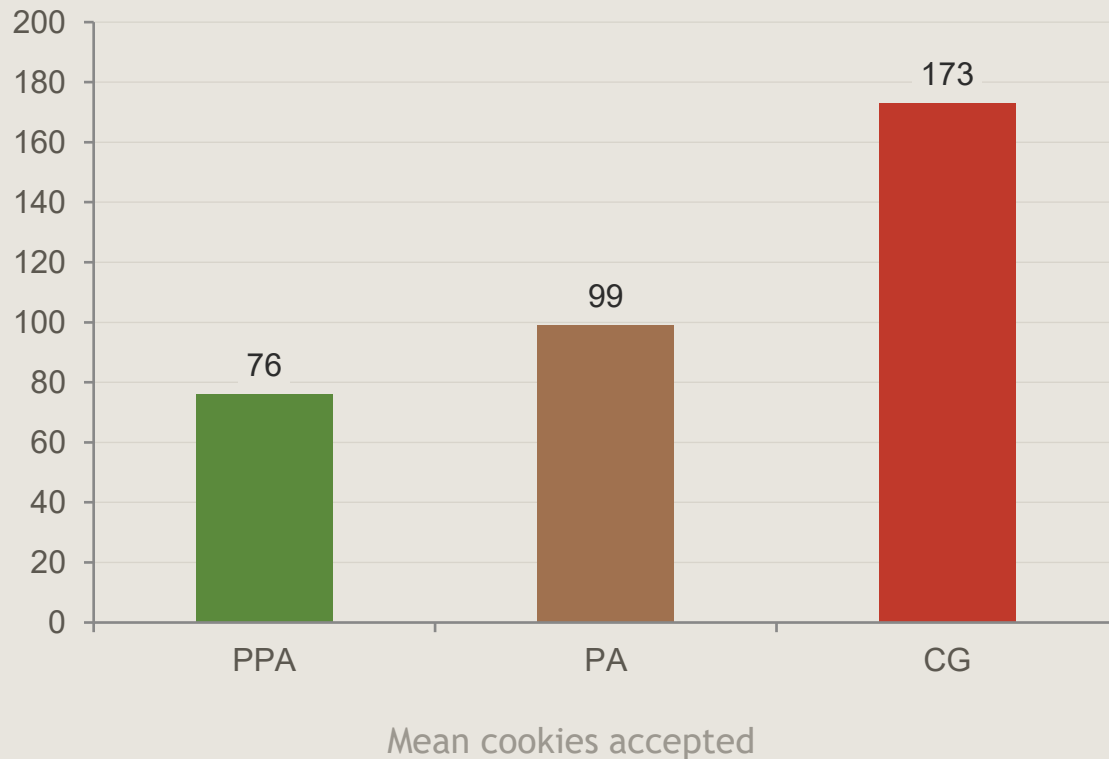
▼ What are third-party cookies?

Functional Cookies	Functional cookies are used to enable the functions of the website. The website cannot function properly without these cookies. According to the law, the website can use them without requiring your consent. For all other types of cookies, your permission is required. ▼ Example
Marketing Cookies	Marketing cookies are used to analyse interactions with websites. This information is then used to tailor advertisements to you. Ads that are personalised in this way are more likely to be clicked on, making them more valuable to advertisers. These cookies are often third party cookies. Third parties may combine information from different sources to build a more accurate picture of your preferences. You will still see ads without marketing cookies. However, their content will not be tailored to you.
All Cookies	This option includes all cookies that the website wishes to set. This means that both functional and marketing cookies are included. It may also include cookies in the categories of preference cookies, statistical cookies, social media cookies and others. This includes third-party cookies. ▼ Example

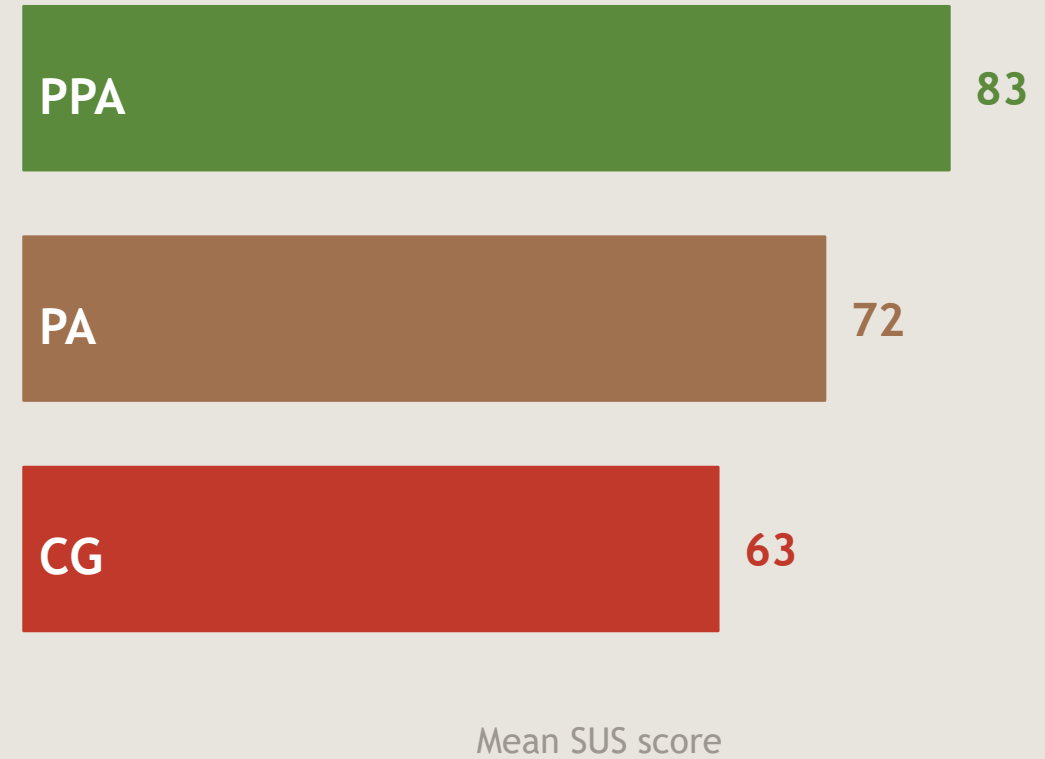
Key Results - We Will Reproduce These in R

Two dependent variables - (1) Nr. of accepted cookies and (2) perceived usability

DV 1: Cookies Accepted (65 - 259)



DV 2: Usability / SUS Score (0 - 100)



Both differences statistically significant · Personalisation has a specific additional effect

3

SECTION 3

FROM RESEARCH QUESTIONS TO HYPOTHESES

The logic of hypothesis-driven research

1

Motivation

2

Paper Pitch

3

Hypotheses

4

Study Design

5

Descriptive Stats

6

Inference Stats

7

Reporting

8

Exercise

Four Concepts You Need First

IV Independent Variable

IV

What you manipulate or vary
The 'cause' in your study

Paper: Banner type: CG / PA / PPA

DV Dependent Variable

DV

What you measure as an outcome
The 'effect' in your study

Paper: Cookies accepted; SUS score

Ho Null Hypothesis

Ho

The default: no effect exists
This is what the test tries to disprove

Paper: No difference in accepted cookies between 3 groups

H₁ Alternative Hypothesis

H₁

Your prediction: an effect exists
Directional or non-directional

Paper: At least one group differs in accepted cookies

From Research Question to Statistical Hypothesis

Research Question

Can privacy knowledge-based personalisation of cookie banners improve users' cookie choices?

Operationalise

IV: Banner type (3 levels: CG / PA / PPA) - randomly assigned
Privacy Knowledge: Measured via OPLIS scale
Personalisation: Cookie explanation texts tailored to privacy knowledge
DV 1: Total cookies accepted (min 65 - max 259)
DV 2: SUS score (0 - 100)

Statistical Hypothesis

H₀: No difference in nr. of accepted cookies (and SUS scores) across the 3 groups
H₁: At least one group differs in cookie choices and SUS scores

SECTION 4

STUDY DESIGN & DATA COLLECTION

4

Variables, scales, sampling, data quality

1

Motivation

2

Paper Pitch

3

Hypotheses

4

Study Design

5

Descriptive Stats

6

Inference Stats

7

Reporting

8

Exercise

Measurement Scales Determine Your Analysis

Categorical

Nominal

Categories, no order
e.g. country, eye color

Paper:

Group: CG / PA / PPA
Gender: male / female / diverse

Use: Mode, counts, chi-square

Ordinal

Ordered, unequal spacing
e.g. grades, clothing size

Paper:

Education level (1-8)
Privacy knowledge:
low/med/high

Use: Median, Kruskal-Wallis

Continuous

Interval

Equal spacing, no true zero
e.g. temperature, year

Paper:

None

Use: Mean, SD, t-test, ANOVA

Ratio

Equal spacing + true zero
e.g. weight, height

Paper:

Cookies accepted (0 = none)
SUS score (debated)

Use: All statistics

Sampling, Recruitment & Data Quality

Sampling in the Paper

Data collection approach

Prolific (crowdsourcing, higher quality than MTurk)

diverse age, gender, education

N = 157 (power-calculated: $d = .5$, $\alpha = .05$, $1-\beta = .80$)

€3 compensation for 15 min - ethical requirement

Limitation: Prolific workers skew tech-savvy → 72% fell in 'high' OPLIS group

Data Quality Considerations

Social desirability bias

Participants may behave more privately when observed. Mitigated by online, unsupervised setting.

Ecological validity

Simulated websites $\neq$ real browsing pressure

Demand characteristics

Participants hypothesize about the goal of the study and aim to behave accordingly → 'good' participant

SECTION 5

DESCRIPTIVE STATISTICS + R

Understanding your data before testing anything

Please refresh the R script
in case it timed-out
<https://bit.ly/cookie-stats>

1

Motivation

2

Paper Pitch

3

Hypotheses

4

Study Design

5

Descriptive Stats

6

Inference Stats

7

Reporting

8

Exercise

Backup: Getting started with Posit Cloud

EXERCISE

1. Download zip “r_files_cookie_analysis” from Moodle
2. Go to posit.cloud → create free account (email + password)
3. New Project → New RStudio Project
4. Files panel (bottom right) → Upload → select ZIP file
5. If only one file shows up: change directory back and forth (target directory: Cloud>project)
6. Run “install.R”
7. Open analysis.R → run code with Ctrl+Enter (Mac: Cmd+Enter)

Descriptive Statistics - What and Why

What descriptives do:

Summarise your sample - they do NOT generalise to populations

Always compute descriptives FIRST - before any test

Catch data errors, understand distributions, tell the story

Key measures:

Mean

Average — sensitive to outliers

Median

Middle value — robust to skew

SD

Spread around the mean

IQR

Spread around the median

Data Preparation – Before Any Analysis

DESCRIPTIVE STATS + R

The dataset has two header rows (common in Survey exports). Here's how to handle it:

	N	O	P	Q	R	S
1	GO02	oplis_score	GO07	GO08_01	SU01_01	SU01_02
2	experimental group	oplis score	privacy knowledge	number of accepted cookies	sus1	sus2
3	PPA	4	low	225	4	4
4	CG	4	low	236	2	3
5	PA	6	low	148	5	1
6	PPA	8	low	70	3	3
7	CG	8	low	72	5	1
8	PPA	8	low	65	4	2
9	PA	8	low	65	4	2
10	PA	9	low	143	1	5
11	PA	9	low	147	4	2
12	PA	10	medium	65	2	4
13	CG	10	medium	238	4	1
14	CG	10	medium	231	3	5
15	PA	10	medium	249	3	4
16	PPA	10	medium	70	3	2
17	PA	11	medium	164	3	4

Data Preparation — Before Any Analysis

The dataset has two header rows (common in Survey exports). Here's how to handle it:

```
library(tidyverse)

# Load real dataset — skip first technical header row
df_raw <- read_excel('cookie_study_data.xlsx',
                    sheet = 'ppa_exp_with_varexplanations',
                    skip = 1) # skip row 1, use row 2 as header

# Rename to clean snake_case
df <- df_raw |>
  select(experimental_group = `experimental group`,
         privacy_knowledge = `privacy knowledge`,
         cookies_accepted = `number of accepted cookies`,
         sus_score = `sus score`,
         age, gender, education,
         ati_score = `ati score`,
         oplis_score = `oplis score`) |>
  mutate(
    cookies_accepted = as.numeric(cookies_accepted),
    experimental_group = factor(experimental_group,
                               levels = c('PPA', 'PA', 'CG'))
  )
```

R: Descriptive Statistics

Follow along!

```
# Descriptive stats by group
df |>
  group_by(experimental_group) |>
  summarise(
    n = n(),
    mean_cookies = round(mean(cookies_accepted, na.rm=TRUE),
1),
    median_cookies = median(cookies_accepted, na.rm=TRUE),
    sd_cookies = round(sd(cookies_accepted, na.rm=TRUE), 1),
    mean_sus = round(mean(sus_score, na.rm=TRUE), 1)
  )
```

```
# Frequencies
table(df$experimental_group)
table(df$privacy_knowledge)

# Any missing?
colSums(is.na(df))
```

R: Visualising Distributions

Always look at the data before running any test

```
# Boxplot - group comparison
ggplot(df, aes(x = experimental_group,
               y = cookies_accepted,
               fill = experimental_group)) +
  geom_boxplot(alpha = 0.8, outlier.colour = 'grey40') +
  scale_fill_manual(values = c(
    'PPA' = '#5B8A3C', 'PA' = '#A0714F', 'CG' = '#C0392B')) +
  labs(title = 'Cookies Accepted by Group',
       x = 'Group', y = 'Cookies') +
  theme_minimal() + theme(legend.position = 'none')
```

```
# Histogram per group
ggplot(df, aes(x = cookies_accepted,
               fill =
experimental_group)) +
  geom_histogram(bins = 20,
alpha=0.8) +
  facet_wrap(~experimental_group) +
  theme_minimal()

# Bar chart - means with error bars
df |> group_by(experimental_group)
|>
  summarise(m =
mean(cookies_accepted),
            se =
sd(cookies_accepted)/sqrt(n())) |>
  ggplot(aes(x=experimental_group,
y=m,

fill=experimental_group)) +
  geom_col() +
  geom_errorbar(aes(ymin=m-se,
ymax=m+se), w=0.2) +
  theme_minimal()
```

5 min break

6

SECTION 6

INFERENCE STATISTICS + R

From description to testing

1

Motivation

2

Paper Pitch

3

Hypotheses

4

Study Design

5

Descriptive Stats

6

Inference Stats

7

Reporting

8

Exercise

Descriptive vs. Inferential Statistics



DESCRIPTIVE

*What does
my sample look like?*

PPA mean = 73.7 cookies
CG median = 231 cookies
True only for this sample.



INFERENCEAL

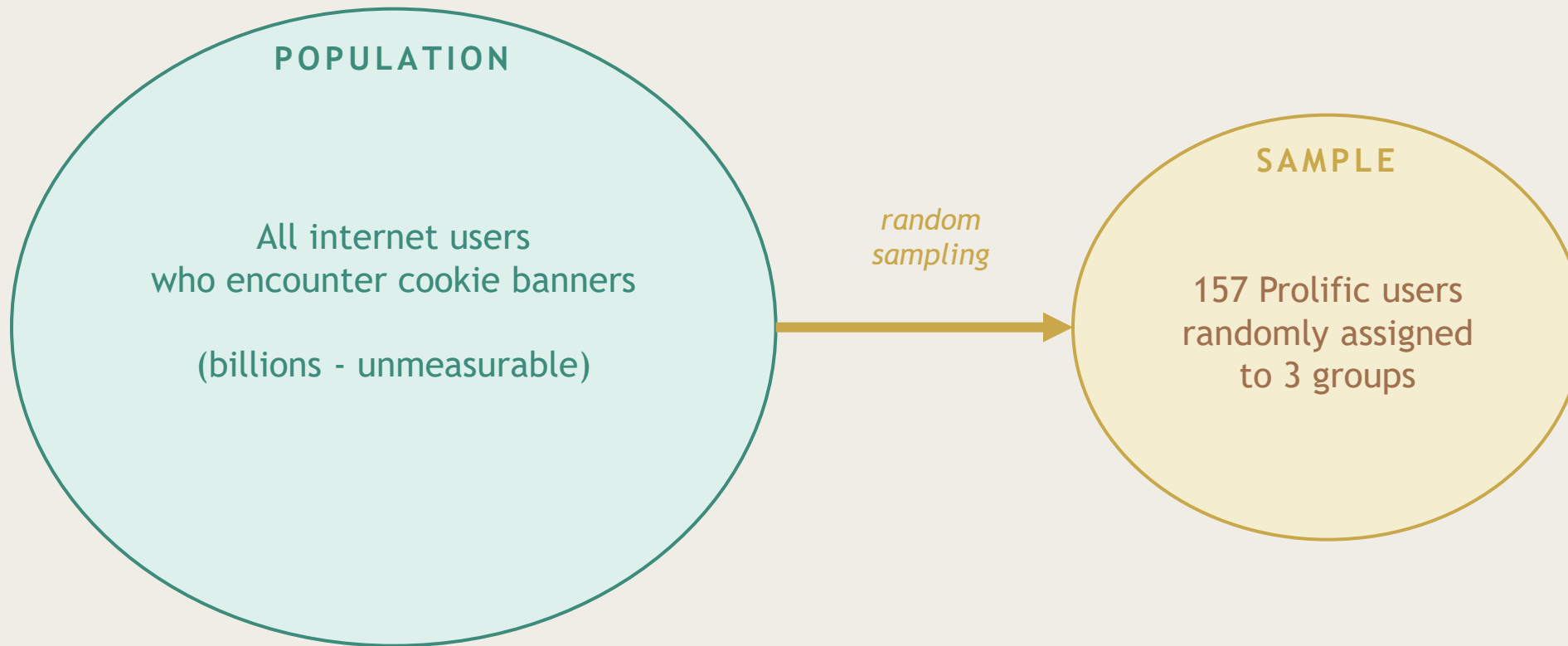
*Is this difference
real in the population?*

$p < .001 \rightarrow$ extremely unlikely under H_0
 $\Rightarrow$ We reject the null
 $\Rightarrow$ The effect is real beyond our sample

Why Do We Need Inference At All?

INFERENCE STATS + R

We can't measure every internet user. We measured 157. So how can we claim anything beyond those 157 people?



The key logic

If the sample was randomly drawn, it is approximately representative.

So a big difference in the sample likely reflects a real difference in the population.

But how likely?
That's what inference tells us.

The Logic Behind Every Statistical Test

INFERENCE STATS + R

1 Assume H_0 is true

Start by assuming there is NO real effect in the population.

For the cookie study: all three banner types lead to the same number of accepted cookies.

2 Ask: how likely is our data under H_0 ?

If H_0 were true and we ran this experiment 10,000 times, how often would we see a difference in accepted cookies between the three groups as large (or larger) as the one we observed?

3 That probability is the p-value

$p = .05$ means: only 5% of random experiments would produce this (or more extreme) result(s) if H_0 were true. Rare enough $\rightarrow$ we reject H_0 .

4 Conclude — carefully

We do NOT prove H_1 . We only say H_0 is implausible given the data.

Always add effect size: statistical significance $\neq$ practical importance.

The p-value: What It Actually Means

p = probability of observing data this extreme (or more)
IF the null hypothesis H_0 were true

Common Misconceptions

✗ $p < .05$ proves the effect is real

✓ p just says this result would be rare under H_0 . It says nothing about effect size.

✗ $p = .04$ and $p = .06$ are fundamentally different

✓ .05 is a convention. Both are weak evidence. Report effect sizes alongside.

✗ Non-significant = H_0 is true

✓ Absence of evidence $\neq$ evidence of absence. Could be underpowered.

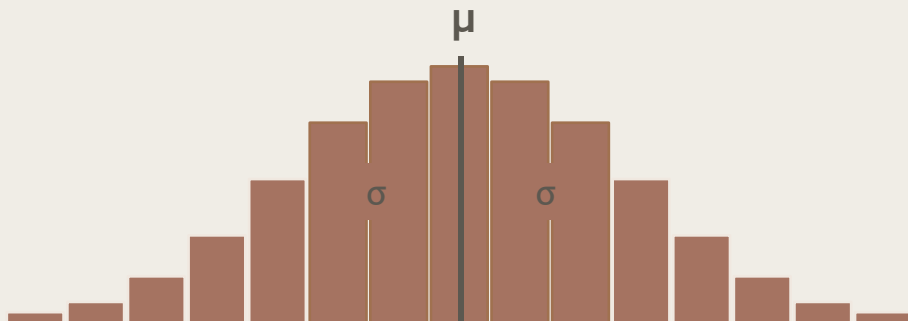
The Normal Distribution - Why It Matters

What it is

A bell-shaped distribution where:

- Most values cluster around the mean
- Values get rarer towards the extremes
- It is symmetric
- Fully described by mean (μ) and SD (σ)

Many natural phenomena approximate it: height, reaction time, measurement error.



Why it matters for inference



Parametric tests (t-test, ANOVA) assume normality

They use the mean and SD, which are only meaningful summaries if the data is approximately normal



If data is NOT normal — use non-parametric tests

These use ranks instead of raw values, so they don't care about distribution shape. Kruskal-Wallis is the 3-group example.

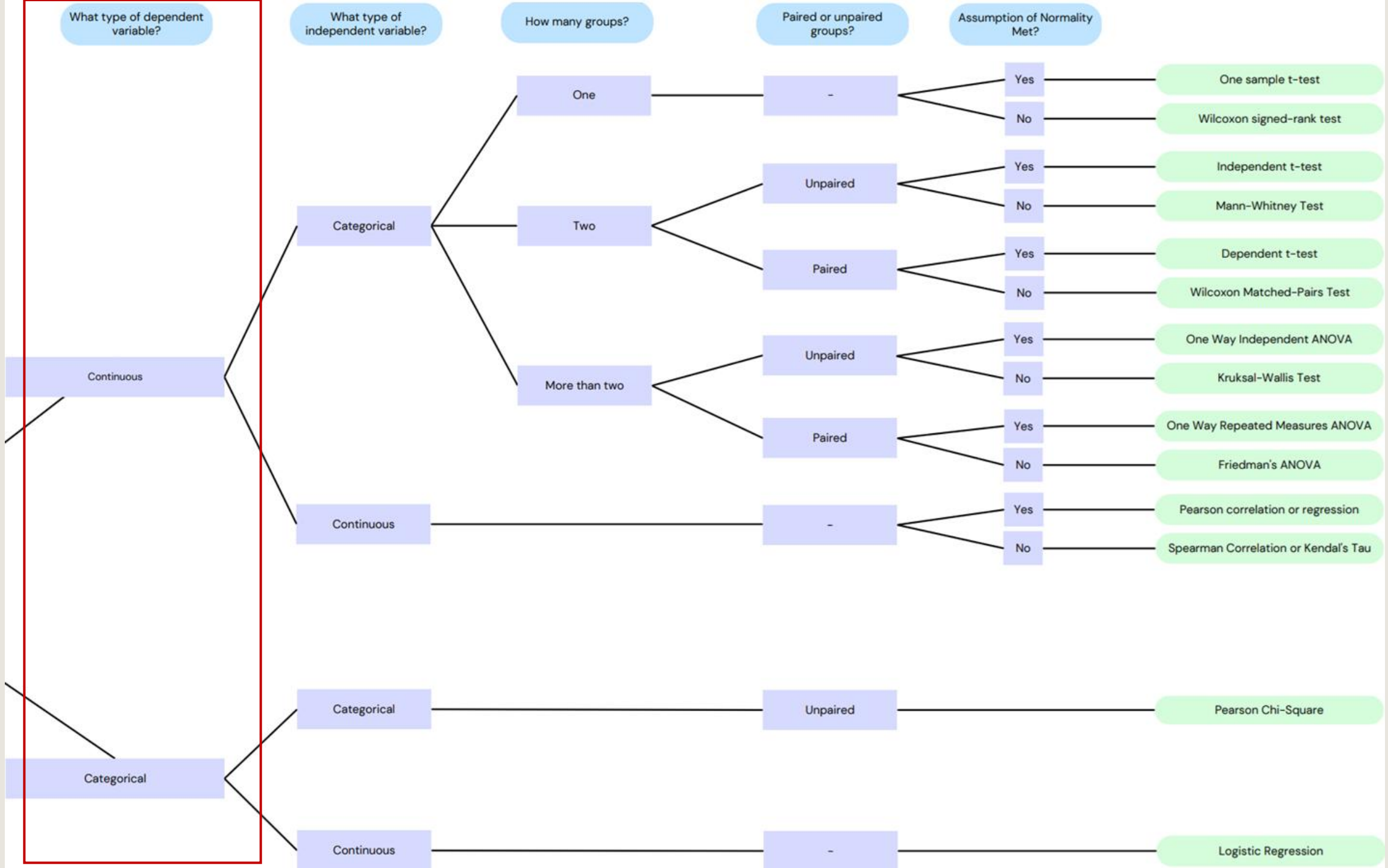


Always check before choosing a test

Visual inspection and `shapiro.test()` per group. If $p < .05 \rightarrow$ non-normal $\rightarrow$ consider using non-parametric. Caution: `shapiro.test()` alone can be insufficient

Choose the correct statistical test

INFERENCE STATS + R



Correlation: Pearson & Spearman

Use case in the cookie study:

Made-up use case: Is there a relationship between ATI score (tech affinity) and cookies accepted? Do tech-savvy users accept fewer cookies?

```
# Pearson (assumes normality)
cor.test(df$ati_score,
         df$cookies_accepted,
         method = 'pearson')

# Spearman (non-parametric, use for
# ordinal or non-normal data)
cor.test(df$ati_score,
         df$cookies_accepted,
         method = 'spearman')

# Scatter plot
ggplot(df, aes(x=ati_score,
               y=cookies_accepted)) +
  geom_point(alpha=0.5) +
  geom_smooth(method='lm') +
  theme_minimal()
```

Interpret:

r = correlation coefficient (-1 to +1)

$p < .05$ = significant association

Positive r = both increase together

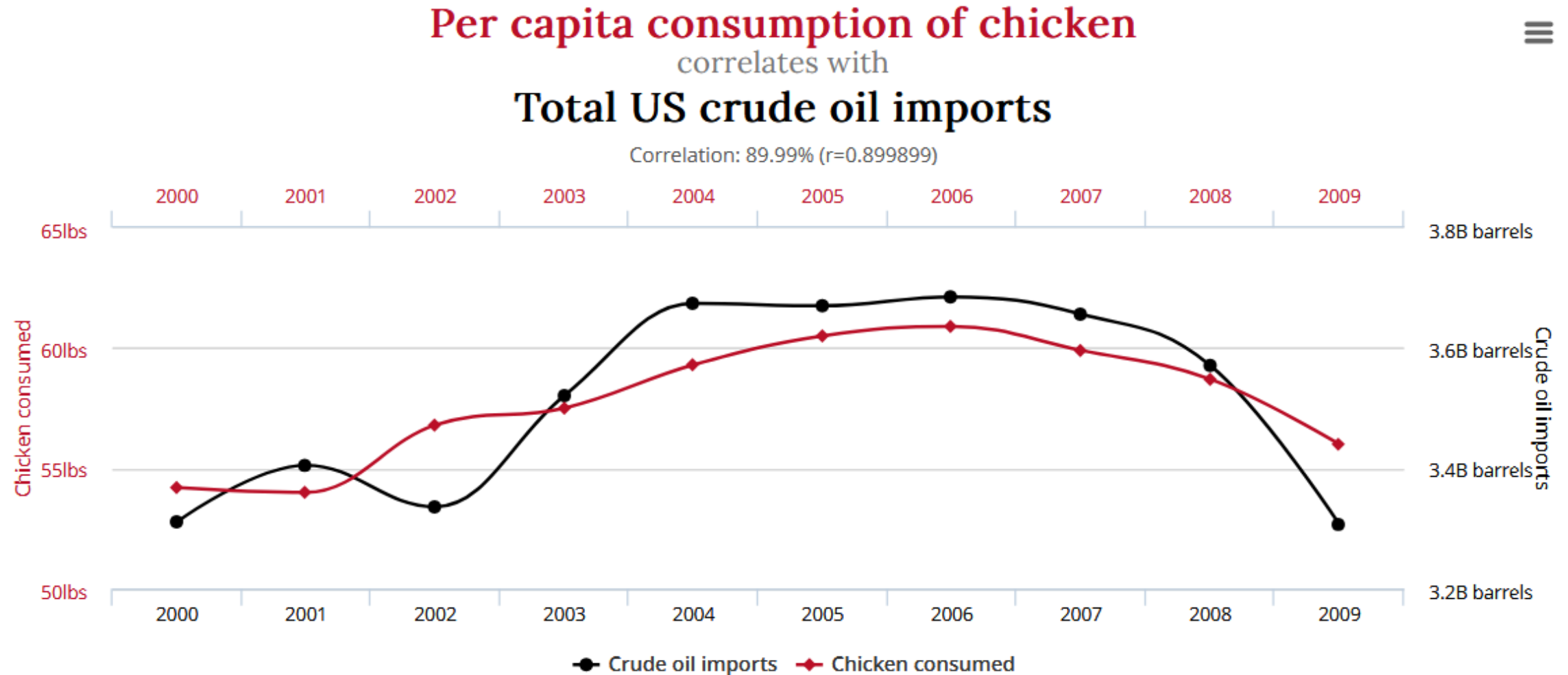
Negative r = one up, other down

Pearson: assumes both variables are normal

Spearman: uses ranks - more robust

! Correlation is not Causation !

INFERENCE STATS + R



Data sources: U.S. Department of Agriculture and Dept. of Energy

tylervigen.com

! Correlation is not Causation !

There is a correlation between ice cream sales and drowning - Why?

X unjustified



A causes B

Ice cream sales cause drowning?

The most tempting - and usually wrong - interpretation.

X unjustified



B causes A

Drowning causes ice cream sales?

Reverse causation. Equally unjustified from correlation alone.

✓ possible



C causes both

Hot summer weather → both ice cream AND swimming

A confounding variable explains both. The real answer, here.

Even though correlations are real, causal stories are often not.

Always ask: what - if any - third variable could explain both?

Linear Regression

Use case in the cookie study:

Made-up use case: Can we predict cookies accepted from OPLIS score and ATI score? Does privacy knowledge independently predict cookie choices, controlling for tech affinity?

```
# Simple linear regression
lm_simple <- lm(cookies_accepted ~ oplis_score,
               data = df)
summary(lm_simple)

# Multiple regression
lm_multi <- lm(cookies_accepted ~
              oplis_score + ati_score + age,
              data = df)
summary(lm_multi)

# Standardised coefficients
library(effects)
standardize_parameters(lm_multi)
```

Interpret:

Output includes:

β (slope): change in DV per unit increase in predictor

R^2 : proportion of DV variance explained by the model

p per predictor: is this variable a significant predictor?

Multiple regression: each β is the effect CONTROLLING for all other predictors

Independent Samples t-test

Use case in the cookie study:

Made-up use case: Do male and female participants differ in their SUS score ratings of the cookie banner? (2 independent groups, 1 continuous DV)

```
# Independent t-test
# (assumes normality per group)
t.test(sus_score ~ gender,
       data = df |> filter(gender %in%
                           c('male', 'female')),
       var.equal = FALSE) # Welch default

# Check normality first
df |> group_by(gender) |>
  summarise(
    sw_p = shapiro.test(sus_score)$p.value
  )

# Effect size (Cohen's d)
library(effectsiz)
cohens_d(sus_score ~ gender, data=df)
```

Interpret:

t-statistic: standardised difference between group means

$p < .05$: groups differ significantly

Cohen's d:

Small = 0.2

Medium = 0.5

Large = 0.8

If normality violated → use Mann-Whitney U

One-Way ANOVA

Use case in the cookie study:

Made-up use case: Do participants across the three education levels (low / medium / high education) differ in their ATI score? (3+ groups, normally distributed continuous DV)

```
# Recode education to 3 levels
df <- df |> mutate(edu3 = case_when(
  education <= 3 ~ 'low',
  education <= 5 ~ 'medium',
  TRUE          ~ 'high') |>
  factor(levels=c('low', 'medium', 'high'))))

# One-way ANOVA
aov_res <- aov(ati_score ~ edu3, data=df)
summary(aov_res)

# Post-hoc (if significant)
TukeyHSD(aov_res)

# Effect size eta-squared
library(effectsize)
eta_squared(aov_res)
```

Interpret:

F-statistic: ratio of between-group variance to within-group variance

$p < .05$: at least one group differs

Post-hoc (TukeyHSD) tells you WHICH pairs

Effect size η^2 (eta-squared):

Small = .01 / Medium = .06 / Large = .14

If normality violated → Kruskal-Wallis

Kruskal-Wallis + Dunn - The Paper's Main Test

This is what the paper actually uses — let's run it on the real data

```
library(rstatix)

# Normality check (expect p < .05 = non-normal)
df |> group_by(experimental_group) |>
  summarise(sw_p = shapiro.test(cookies_accepted)$p.value)

# Kruskal-Wallis test
kw <- kruskal.test(cookies_accepted ~ experimental_group,
                  data = df)
print(kw) # chi-squared = 43.7, df=2, p < .001

# Effect size
df |> kruskal_effsize(cookies_accepted ~ experimental_group)

# Post-hoc: Dunn's test with BH correction
dunn <- df |> dunn_test(cookies_accepted ~ experimental_group,
                      p.adjust.method = 'BH')

# Add effect size r = |Z| / sqrt(N)
dunn |> mutate(r = abs(statistic) / sqrt(n1 + n2))
```

Kruskal-Wallis rank sum

```
chi-sq = 43.725
  → large test statistic
df = 2 (groups - 1)
p < 0.001
  → reject H0
```

Dunn post-hoc (BH):

```
PPA vs PA: Z=2.15, p=.03
  r = .19 [small-medium]
PPA vs CG: Z=6.51, p<.001
  r = .52 [large]
PA vs CG: Z=4.20, p<.001
  r = .37 [medium]
```

Doing It Right - Scientific Integrity



α -level (Type I Error Rate)

Set to 0.05 before data collection.
Tolerate a 5% chance of false positive.
This is a convention, not a law - some fields use 0.01.



Statistical Power, Optimal Sample Size

Set to 0.80 = 80% chance of detecting a true effect.
Low power → miss real effects (false negatives).
The paper: $d=.5$, $\alpha=.05$, power=.80 → $N=157$ (G*Power).



HARKing - Don't Do It

Hypothesising After Results are Known.
Looks like confirmatory science, but it isn't.
Inflates false positives. Pre-registration prevents it.
Always formulate H_0/H_1 before collecting data.



p-Hacking, Correction for Multiple Comparisons

Running many tests, reporting only the significant one.
BH / Bonferroni corrections help when running multiple tests legitimately.
Always report all tests, not just the 'good' ones.

Multiple Comparisons — Why We Correct

Running 3 tests at $\alpha = .05$ gives a 14% chance of at least one false positive

P(at least one false positive):



Bonferroni

Divide α by number of tests
Very conservative — increases false negatives

Benjamini-Hochberg (BH)

Controls false discovery rate
Less conservative, widely used in HCI research
→ Used in the paper

7

SECTION 7

REPORTING RESULTS

From R output to published sentence

1

Motivation

2

Paper Pitch

3

Hypotheses

4

Study Design

5

Descriptive Stats

6

Inference Stats

7

Reporting

8

Exercise

APA Reporting Template

Turn R output into a publishable sentence – step by step

Reported result – cookies accepted

Overall test:

"A Kruskal-Wallis test revealed a significant difference in the number of accepted cookies between the three groups, $\chi^2(2) = 43.73$, $p < .001$."

Post-hoc:

"Post-hoc pairwise comparisons (Dunn's test, BH correction) showed that the PPA group (Mdn = 65) accepted significantly fewer cookies than both the PA group (Mdn = 65; $Z = 2.15$, $p = .03$, $r = .19$) and the control group (Mdn = 231; $Z = 6.51$, $p < .001$, $r = .52$). The PA group also accepted significantly fewer cookies than the control group ($Z = 4.20$, $p < .001$, $r = .37$)."

Effect size:

"Effect sizes ranged from small-medium ($r = .19$) to large ($r = .52$).

✓ Test name +
statistic

✓ df

✓ p-value

✓ Group medians

✓ Effect size r

✓ Correction
method

Complete R Script – Take This Away

Runs top-to-bottom on the real dataset – also in your cloud project as `kruskal_wallis_complete.R`

```
# 0. Libraries
library(tidyverse); library(readxl)
library(rstatix); library(effectsize)

# 1. Load real data (two-row header → skip=1)
df <- read_excel('cookie_study_data.xlsx',
  sheet='ppa_exp_with_varexplanations', skip=1) |>
  select(experimental_group='experimental group',
    cookies_accepted='number of accepted cookies',
    sus_score='sus score', age, gender,
    ati_score='ati score', oplis_score='oplis
score',
    privacy_knowledge='privacy knowledge') |>
  mutate(cookies_accepted=as.numeric(cookies_accepted),
    experimental_group=factor(experimental_group,
      levels=c('PPA', 'PA', 'CG'))))

# 2. Descriptives + plots
df |> group_by(experimental_group) |>
  summarise(n=n(), mean=mean(cookies_accepted),
    median=median(cookies_accepted))
ggplot(df, aes(experimental_group, cookies_accepted,
  fill=experimental_group)) +
  geom_boxplot() + theme_minimal()

# 3. Normality check
df |> group_by(experimental_group) |>
  summarise(sw=shapiro.test(cookies_accepted)$p.value)

# 4. Kruskal-Wallis + effect size
kruskal.test(cookies_accepted~experimental_group, df)
df |>
  kruskal_effsize(cookies_accepted~experimental_group)

# 5. Post-hoc + r
df |> dunn_test(cookies_accepted~experimental_group,
  p.adjust.method='BH') |>
  mutate(r = abs(statistic)/sqrt(n1+n2))
```

SECTION 8

EXERCISE

Now you try it — independently

Caution: session times out
after 10min of inactivity!
<https://bit.ly/cookie-stats>

1

Motivation

2

Paper Pitch

3

Hypotheses

4

Study Design

5

Descriptive Stats

6

Inference Stats

7

Reporting

8

Exercise

Exercise: Replicate the SUS Analysis (15min)

EXERCISE

Same pipeline, different outcome. Work individually or in pairs.

1

Compute group descriptives for `sus_score` (mean, median, SD). Which group rated usability highest?

💡 *Copy your summarise() code, replace `cookies_accepted` with `sus_score`*

2

Create a boxplot of `sus_score` by `experimental_group`.

💡 *Copy your ggplot code, change y variable*

3

Check normality of `sus_score` per group. What do you conclude?

💡 *`shapiro.test(sus_score)` per group*

4

Run the appropriate test to compare SUS scores across all 3 groups.

💡 *`kruskal.test(sus_score ~ experimental_group, df)`*

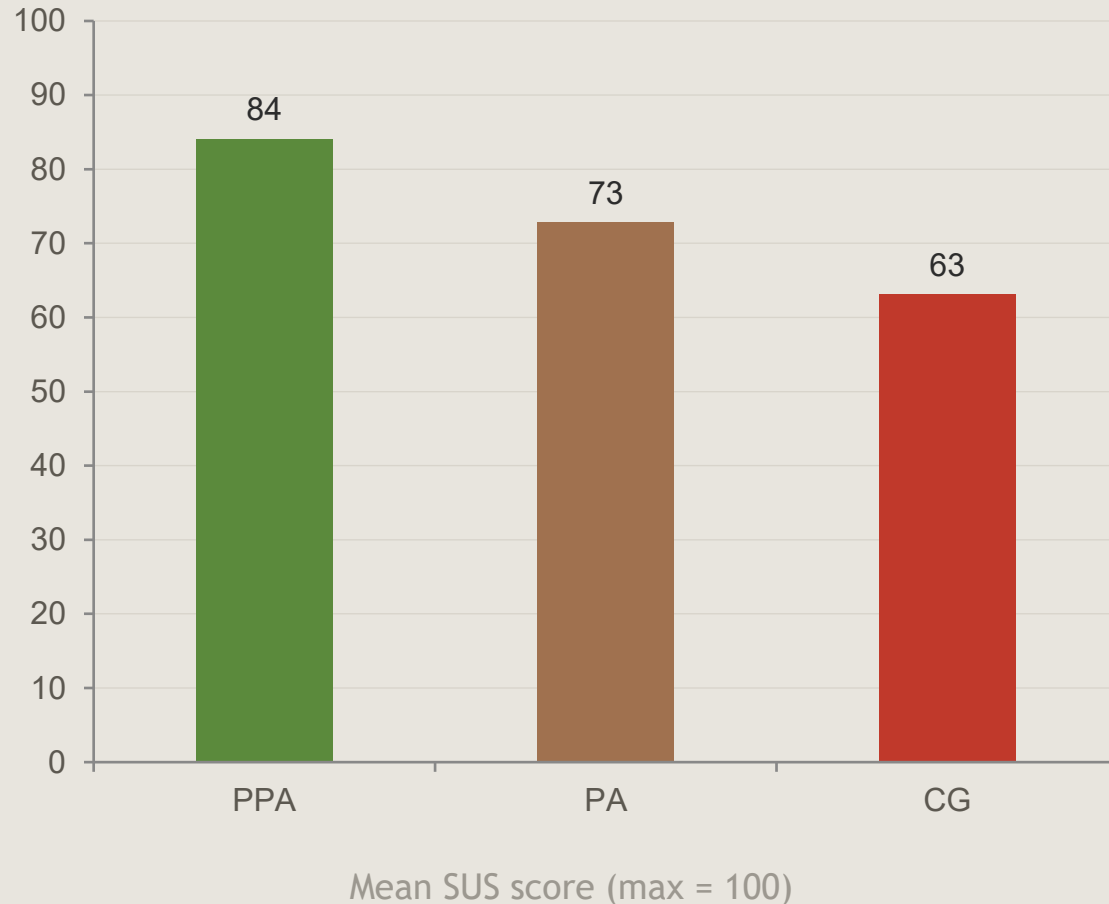
5

Run post-hoc comparisons. Which pairs differ? Write one result sentence using the APA template.

💡 *`dunn_test(..., p.adjust.method='BH')` + add r effect size*

Exercise: Expected Results

EXERCISE



Key Numbers

Kruskal-Wallis: $\chi^2(2) = 29.80, p < .001$

Dunn (BH): all 3 pairs significant

- PPA vs PA: $Z = -3.12, p < .01, r = .27$
- PPA vs CG: $Z = -5.43, p < .001, r = .47$
- PA vs CG: $Z = -2.18, p = .03, r = .19$

Conclusion: PPA highest usability, CG lowest.
Personalisation improves both privacy choices AND user experience.

"A Kruskal-Wallis test revealed a significant difference in SUS scores across groups, $\chi^2(2) = 29.80, p < .001$. Post-hoc tests (Dunn, BH) showed all three groups differed significantly, with PPA rated highest ($M = 83$) and CG lowest ($M = 63$)."

